Ngo Chi Nghia

Fresher AI Engineer | GenAI, LLMs & RAG Systems

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Summary

Proactive and detail-oriented **AI Engineer** with solid hands-on experience in **Generative AI, Large Language Models (LLMs), RAG architectures, and AI Agents**. Proven background in developing enterprise-grade AI chatbots, hybrid retrieval systems, and scalable RESTful AI backends using **Python, LangChain, LangGraph, FastAPI, and ChromaDB**. Experienced in containerized deployment (Docker/Linux) and optimizing model latency, accuracy, and operational token costs.

Technical Skills

Core Programming: Python (Advanced), SQL, C/C++
GenAI & LLM Frameworks: LangChain, LangGraph, AI Agent workflows, RAG Pipelines, Prompt Engineering, OpenAI API, Azure OpenAI (familiarity), Gemini, Groq
Vector DBs & Retrieval: ChromaDB, Vector Embeddings (multilingual-E5), Hybrid Search (Dense + BM25), Reciprocal Rank Fusion (RRF)
Machine Learning & CV: TensorFlow, Scikit-learn, YOLOv8, OpenCV, Hugging Face
Backend & Integration: FastAPI, RESTful APIs, Enterprise System & Data Integration
DevOps & Cloud: Docker, Docker Compose, Nginx, Linux (Ubuntu), Git
AI-Assisted Dev: Claude Code, OpenAI Codex, LLM-assisted Coding, Debugging & Refactoring

Experience

Tien Phong Engineering Technology Co., Ltd. Can Tho, Vietnam
AI Engineer Intern *Mar. 2026 – Jul. 2026*

- Developed scalable **RESTful AI microservices** using **FastAPI** to power enterprise LLM workflows and Chatbot applications.
- Designed and optimized **Retrieval-Augmented Generation (RAG)** pipelines with **ChromaDB** and **multilingual-E5** embeddings, significantly improving query response time and retrieval relevance.
- Integrated **OpenAI**, Gemini, and Groq APIs with tailored prompt engineering to balance model accuracy and token consumption costs.
- Containerized and deployed AI services using **Docker & Docker Compose** on Ubuntu Linux servers for reliable staging and production pipelines.
- Collaborated with software engineering teams to integrate AI models smoothly into existing enterprise databases and production systems.

Selected Projects

PolyAqua Agent – Multi-modal AI Assistant & RAG | GitHub Repo | Portfolio 2026

- Engineered an end-to-end AI assistant combining multi-modal disease diagnostics, aquaculture calculators, and RAG-based advisory.
- Built a hybrid dense-sparse retrieval pipeline (**ChromaDB + BM25** with **RRF fusion**) across domain technical literature to eliminate hallucinations.
- Trained a custom **YOLOv8n-cls** model for aquaculture disease classification and integrated OpenAI models for contextual recommendations.

Vietnamese Legal Document RAG Web App | GitHub Repo | Portfolio Jan. 2026

- Deployed an enterprise-style legal QA Chatbot application utilizing **LangGraph** for multi-step reasoning and grounded prompt validation.
- Implemented hybrid retrieval over Vietnamese IT/cybersecurity legal documents, delivering source-cited answers with high precision.
- Containerized frontend and backend services using **Docker Compose** and Nginx on Ubuntu server.

Energy-Based Retrieval Clustering RAG (ERC-RAG) | GitHub Repo | Portfolio Jan. 2026

- Proposed and evaluated a cluster-level retrieval method over 120,165 ChromaDB vectors to optimize retrieval latency and accuracy on large-scale datasets.
- Achieved high benchmark results: **NDCG@5 0.87**, **Cosine 0.96**, **ROUGE-2 F1 0.74**, and **BLEU-2 0.71** in multi-query evaluations.

Publications & Awards

Co-authored 2 Research Papers on RAG Architecture Optimization:

- *"ERC-RAG: An Energy-based Retrieval Clustering for RAG in Vietnamese Economy Chatbots"* | Paper
- *"Multi-Query ERC-RAG: An Energy-based Retrieval Clustering Framework for Retrieval-Augmented Generation"*

Finalist Certificate – FPTU Can Tho Olympic AI 2026 (University-level Artificial Intelligence Competition)

Education

Nam Can Tho University (NCTU) — Can Tho, Vietnam

Engineer in Information Technology (GPA: 3.44 / 4.0) — *Oct. 2022 – Aug. 2026*

Languages

English: VSTEP B1 | **Vietnamese:** Native